Warming Switzerland: supporting our community to thrive in a hotter future

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People in Switzerland are learning to live with a warming climate. We now experience 10-15 hot days per year, up from just five in 1990. The 2003 European heatwave, which increased Swiss mortality by 1.5% in that year, showed how quickly intense heat can impact our communities. Living in hotter cities and sleeping through tropical nights can take a high toll on quality of life, particularly for older people and other vulnerable groups. Yet unlike floods or storms, heat is often less visible, even as it influences other risks and changes the way we live and work. Adapting to this warmer future is a shared effort, bringing together communities, businesses, public authorities and the re/insurance sector to help Switzerland remain resilient and thrive.

Switzerland learns to live with the heat

From its cities to its mountain peaks, Switzerland is warming faster than much of the rest of the world. Our number of hot days, when temperatures rise above 30°C, has already increased to about 10–15 per year, up from just five in 1990, on average.[1] In a 3°C warming scenario, that number could rise to about 20–30 per year by the middle of this century.[2]

For many of us, these changes are becoming easier to notice. We experience more frequent heatwaves, hotter and drier summers, and milder winters with less snowfall. Switzerland's iconic mountain landscape makes it particularly sensitive to warming. High altitudes tend to warm faster, and the loss of snow and ice can speed up this process.[3]

Rising temperatures are affecting many aspects of life here in Switzerland. Hotter cities, warmer nights and more frequent water stress are expected to place growing pressure on the way we live, work, and care for our health and wellbeing. Heat is different from the typical natural hazards that Switzerland experiences, such as floods, hail and storms, which can cause visible and immediate damage. It develops gradually and can be easier to underestimate – even as it amplifies some of these natural hazards.

Switzerland has invested extensively in resilience measures to manage its natural hazard risks. As temperatures continue to rise, adapting to warmer conditions may become an increasingly important part of that resilience story. We are all key to adapting our infrastructure, communities, economy and way of life to this new warmer future. Strengthening resilience is a shared effort, with federal and cantonal authorities, municipalities, utilities, emergency services, insurers, businesses and communities all playing a vital role.

Hot days in Switzerland are becoming more frequent

Days with daily maximum temperature >=30°C, per year

Source: MeteoSwiss and ETH Zürich

Switzerland is getting warmer, faster

Switzerland's average surface temperatures between 2015 and 2024 were about 2.8°C above the pre-industrial baseline (1850-1900). It is warming faster than Europe (2.2°C) and more than twice as fast as the rest of the world (1.2°C).[4] This trend is likely to continue as temperatures rise.[5] The impacts of rising temperatures on key climate and environmental metrics in Switzerland include more frequent and intense heavy rainfall,[6] among other changes. For an overview, see the Appendix.

Rise in average surface temperatures, from pre-industrial levels to 2015-2024, °C

Source: Swiss Academy of Sciences

Heat can take a serious toll on health

The 2003 European heatwave was one of the deadliest events in Europe from any peril in recent decades. It increased mortality in Switzerland by about 1.5% that year,[7] and showed how quickly heat can affect people's health and place added pressure on healthcare systems – especially in a country with an ageing population.

Extreme heat can pose serious health risks, particularly for the elderly and those with existing health conditions. It increases the risk of heatstroke, dehydration, cardiovascular strain and respiratory illness. Yet, unlike losses from floods or storms, the human toll is often less visible and less closely tracked.

This challenge is most visible in cities, where heat stress is the highest. Swiss cities warm faster during the day and cool down more slowly at night, with urban-rural temperature differences reaching up to 6°C.[8] Tropical nights — when temperatures do not fall below 20°C — increased 12-fold in Zurich, 4.5-fold in Basel and 5-fold in Geneva between 1981–2010 and 2011–2025.[9] This trend is expected to continue.[10]

Hot days and especially hot nights can be difficult to manage without access to cooling. Yet air conditioning is installed in only about five percent of Swiss homes,[11] as buildings have traditionally been designed to keep people warm rather than cool. In this context, measures such as heat warnings can help people better prepare, while cooling centres, greener urban planning and better building insulation can help people stay safe and comfortable.

Interplay of winter and summer mortality

Outside of years with extreme heatwaves, the health impacts of a warmer climate can be more complex than they first appear. Studies suggest that increases in heat-related deaths during summer may be partially, or in some regions fully, offset by fewer deaths in winter. In Switzerland's higher-latitude regions, where winter-related mortality has historically exceeded heat-related mortality, this pattern may continue in the near- to medium term.[12] Over the longer term, however, continued warming may lead to increases in summer mortality that could outweigh reductions in winter mortality, with implications for medical, life and workers' compensation claims.[13]

Some risks are familiar – until you turn up the heat

Switzerland is no stranger to natural catastrophes and has a strong track record of building resilience. This includes robust building standards, careful land-use planning, investment in adaptation and high levels of insurance coverage. Against this backdrop, the country's insured losses have grown relatively modestly over the past 30 years. Flood remains Switzerland's main natural catastrophe risk, accounting for around 60% of average annual insured exposure-normalised losses,[14] followed by severe convective storms at about 20% and winter storms at around 11%, data from Swiss Re Institute shows.

Heat, however, can reshape the risks Switzerland is already familiar with. After periods of intense heat, dry soils are often less able to absorb water, increasing the potential for flash flooding when heavy rainfall arrives. Drought can also leave weakened crops more vulnerable to hail damage. Wildfires also typically increase the risk of erosion and debris flows by removing vegetation and changing how soils absorb water when heavy rainfall occurs. Rising temperatures can also destabilise mountain slopes, increasing the risk of ice and rock avalanches.

Blatten rock and ice avalanche

Described by the Swiss Seismological Service as one of the largest mass movements ever recorded in Switzerland,[15] the Blatten rock and ice avalanche of May 2025 resulted in an insured loss of CHF 320 million. It illustrates how chronic change can already influence the risk landscape associated with severe loss events today.

A hot workday can strain the economy

Beyond health impacts, heat can affect the way people work. Outdoor jobs can become harder during extreme heat, potentially requiring shifts to cooler hours or night work, resulting in higher labour costs (+10%).[16] Heat can disrupt schools and public services: after the May 2026 heatwave, for example, the Swiss teachers' association called for classes to be suspended when temperatures reach 30°C.[17]

Agriculture is feeling the effects as well. Heavier rainfall can hinder work in fields, while heat interacts with other climatic shifts. Milder winters and earlier vegetation development have increased exposure to spring frost in fruit production. More frequent droughts reduce crop yields and grass production, particularly in western Switzerland. Adaptation measures such as sprinkler protection, water reservoirs and drip irrigation can reduce losses, but these often require significant investment or regulatory adjustment.

Heat also places pressure on the infrastructure and assets that support daily life, both directly and through wider system stress. It can accelerate wear and tear on roads, runways, rail lines, bridges, buildings and water pipelines that were not designed for a warmer climate. Prolonged heat can raise electricity demand for cooling while reducing generation and transmission efficiency, increasing the risk of grid overload and power shortages.

Thermal stress at Beznau nuclear plant

In June 2025, Axpo reduced output at the Beznau nuclear power plant after the Aare reached 25°C and later shut one unit, while the other operated at 50% capacity,[18] to avoid adding further thermal stress to the river and to comply with environmental requirements.[19]

Creating a Switzerland that can thrive in the heat – together

Heat requires a broader policy response than many traditional hazards, as it affects many systems at once. Rather than treating it as a standalone hazard, adaptation can be woven into many parts of daily life, from planning and infrastructure investment to public health, and economic policy.

Switzerland has already strengthened its preparedness through warnings, heat-health action plans, public awareness campaigns, city heat indicators and targeted protection of vulnerable groups. It is beginning to extend the resilience model used for established natural hazards, based on monitoring, early warning and coordinated response, to emerging heat-related risks.

The next step may be to build on these foundations by embedding heat resilience more systematically in planning, building rules and local delivery, especially for vulnerable facilities and public space. Green infrastructure, unsealed surfaces and sponge-city concepts can lower temperatures while also improving rainwater retention and easing pressure on drainage systems during heavy rainfall. In the Canton of Zurich, Winterthur's sponge-city approach and Regensdorf's Zwhatt project,[20] provide examples of how heat adaptation can be built into urban design and planning.

Much of this resilience is built at the local level. Municipalities are often best placed to turn climate risk into practical action, helping communities prepare for and adapt to changing conditions. The value of local action is clear from cases such as the landslide onto Blatten, and the similarly threatened village of Brienz,[21] where close monitoring and preparedness helped to protect exposed communities.

Insuring a different kind of risk

For insurers, heat extends risk well beyond direct property damage and life and health insurance, into areas such as business interruption, agriculture, casualty and liability. Unlike floods or storms, many of its costs emerge gradually through impacts such as excess mortality, lost working hours, pressure on healthcare systems and disruption when energy, transport and water systems are stretched beyond their limits. This can make heat harder to insure through traditional products, as losses are often spread out and difficult to attribute. Some losses can be transferred, but many of the wider economic and social costs remain difficult to insure at scale.

Insurance responses are evolving. In agriculture, multi-peril crop insurance (MPCI) supports resilience and in 2025, the Swiss federal government began subsidising MPCI premiums to encourage wider adoption, although the protection gap remains high. Alternative risk transfer solutions, such as parametric insurance, may be better suited where heat can be measured through a clear trigger, for example to provide income support for workers exposed to extreme heat.[22] Building resilience to heat is likely to involve collaboration across the public and private sectors. Alongside providing risk transfer, insurers can contribute by sharing insights with public authorities and supporting discussions on adaptation.

Appendix

Key changes in Switzerland's climate and environment by Climatic-Impact Driver (CID) based on observational data, and sectors exposed to impact

Climatic-Impact Driver:	Heat and cold	Snow and ice	Wet and dry
Key climate and environmental changes in Switzerland	**Up to 3x higher** number of days with 30°C and above since 1990	**-65%** in glacier volume since 1850	**+12%** more intense, **+26%** more frequent daily precipitation events since 1901
	-60% in frost days since 1961	**-20% to -50%** number of snow days since 1970	**+20%** more intense heaviest 10-minute summer precipitation since 1980s
	+2 to 4 weeks in vegetation time since 1961	**+300 to 400m** zero-degree limit (altitude at which air temperature is 0°C) since 1961	**-5% to -10%** summer soil moisture since early 1980s
Sectors exposed to impact	Health, water resources, energy demand and supply, agriculture, forestry, transportation, telecommunications, construction, infrastructure	Public safety, hydropower, water resources, transportation, alpine infrastructure and property, winter tourism	Water resources, hydropower, water supply, property, infrastructure, transportation, inland shipping

Source: Swiss Academy of Sciences, MeteoSwiss, Swiss Re Institute

References

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